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$$f(x) = \ln \left(\sqrt[3]{3x + \sqrt{9x^2 + 1}} \right)$$

$$f(x) = \ln \left((3x + \sqrt{9x^2 + 1})^{\frac{1}{3}} \right)$$

$$f(x) = \frac{1}{3} \ln (3x + \sqrt{9x^2 + 1})$$

$$f'(x) = \frac{1}{3} \cdot \frac{1}{3x + \sqrt{9x^2 + 1}} \cdot (3x + \sqrt{9x^2 + 1})'$$

$$(3x + \sqrt{9x^2 + 1})' = 3 + \frac{1}{2\sqrt{9x^2 + 1}} \cdot 18x$$

$$(3x + \sqrt{9x^2 + 1})' = 3 + \frac{9x}{\sqrt{9x^2 + 1}}$$

$$(3x + \sqrt{9x^2 + 1})' = \frac{3\sqrt{9x^2 + 1} + 9x}{\sqrt{9x^2 + 1}}$$

$$(3x + \sqrt{9x^2 + 1})' = \frac{3(3x + \sqrt{9x^2 + 1})}{\sqrt{9x^2 + 1}}$$

$$f'(x) = \frac{1}{3} \cdot \frac{1}{3x + \sqrt{9x^2 + 1}} \cdot \frac{3(3x + \sqrt{9x^2 + 1})}{\sqrt{9x^2 + 1}}$$

$$f'(x) = \frac{1}{\sqrt{9x^2 + 1}}$$

References